Examiner only

4. Colorimetry is a method used to determine the concentration of haemoglobin in blood.

The table below shows how the absorbance depends on the concentration of haemoglobin. One mean value is missing.

Concentration of haemoglobin (mg/100 cm^3)	Absorbance (units)				
	Test 1	**Test 2**	**Test 3**	**Test 4**	**Mean**
50	0.20	0.21	0.20	0.19	0.20
100	0.37	0.41	0.63	0.39	
150	0.60	0.61	0.60	0.60	0.60
200	0.80	0.79	0.81	0.83	0.81
250	0.99	0.99	0.99	0.99	0.99

(a) Rhodri and Dafydd calculated the missing mean absorbance value. Rhodri calculated a mean value of 0.45 and Dafydd calculated a value of 0.39. Show how both values were obtained and explain which is the better method of calculating a mean. [3]

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(b) (i) Use the data in the table to plot a graph of mean absorbance against concentration of haemoglobin on the grid below and draw a suitable line. [4]

(ii) Describe the relationship between absorbance and concentration of haemoglobin. [2]

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....

(c) One part of the structure of haemoglobin has the chemical formula $FeN_4C_6H_{18}O_4$.

(i) Calculate the relative molecular mass (M_r) of this part of haemoglobin. [2]

M_r = ...

(ii) Calculate the total mass of this part of haemoglobin in 2.5 moles. [2]

Mass = ...

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